

Conversion of hardening laws

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Hardening laws $\sigma(\varepsilon)$ and $\tau(\Gamma)$

Hardening laws $\sigma(\varepsilon)$ describe the evolution of the yield stress σ as a function of an equivalent plastic strain ε . Normally the hardening curves are obtained by fitting to tensile test results in a direction for which σ and ε correspond to the tensile stress and the plastic tensile strain.

In polycrystalline plasticity models one uses the concept of a critical resolved shear stress τ . Hardening may be described by a model where τ is as function of the plastic slip Γ . With $\dot{W}$ the rate of work per unit volume, the slip rate $\dot{\Gamma}$ is defined by:

$$\dot{\Gamma} = \frac{\dot{W}}{\tau} \quad (1)$$

The plastic slip is given as:

$$\Gamma = \int \dot{\Gamma} dt \quad (2)$$

Let σ be the yield stress for a tension test in a certain direction (e.g. the rolling direction of a sheet material). Let z be the tensile yield stress divided by τ :

$$z = \frac{\sigma(\tau)}{\tau} \quad z = \text{Taylor factor } M, \text{ only if von Mises used as equivalent strain rate!} \quad M = \frac{P}{\dot{\varepsilon}_o \tau^c} = \frac{\sigma_v : \dot{\varepsilon}}{\dot{\varepsilon}_o \tau^c} \quad (3)$$

The value of z can be calculated by a polycrystalline plasticity model, or the plastic potential that approximates this model. **Set $\tau = 1$, then $z = \sigma$!**

Let D be the plastic strain rate in the tensile direction. The rate of plastic work per unit volume for the tensile test is then given by:

$$\dot{W} = \sigma D \quad (4)$$

It follows from (1), (3) and (4) that:

$$\dot{\Gamma} = z D \quad (5)$$

Assuming constant z (i.e. constant texture), it follows from (2) and (5) that

$$\Gamma = z \varepsilon \quad (6)$$

Where:

$$\varepsilon = \int D dt \quad (7)$$

Equations (3) and (6) can be used to convert $\sigma(\varepsilon)$ hardening laws into $\tau(\Gamma)$.

Conversion of $\sigma(\varepsilon)$ hardening laws into $\tau(\Gamma)$

Swift law with parameters K_s (or σ_0), ε_0 and n_s

$$\sigma(\varepsilon) = K_s (\varepsilon + \varepsilon_0)^{n_s} = \sigma_0 \left(1 + \frac{\varepsilon}{\varepsilon_0} \right)^{n_s} \quad \text{with } \sigma_0 = K_s \varepsilon_0^{n_s} \quad (8)$$

Using (3) and (6) it follows that:

$$\tau(\Gamma) = K'_s (\Gamma + \Gamma_0)^{n'_s} = \tau_0 \left(1 + \frac{\Gamma}{\Gamma_0} \right)^{n'_s} \quad \text{with } \tau_0 = K'_s \Gamma_0^{n'_s} \quad (9)$$

$$K'_s = \frac{K_s}{z^{1+n_s}} \quad (10)$$

$$\tau_0 = \frac{\sigma_0}{z} \quad (11)$$

$$\Gamma_0 = z \varepsilon_0 \quad (12)$$

$$n'_s = n_s \quad (13)$$

The hardening coefficient H is given by:

$$H(\Gamma) = \frac{d\tau}{d\Gamma} = K'_s n'_s (\Gamma + \Gamma_0)^{n'_s-1} = \frac{\tau_0 n'_s}{\Gamma_0} \left(1 + \frac{\Gamma}{\Gamma_0} \right)^{n'_s-1} \quad (14)$$

Voce law with parameters σ_0 , K_v and n_v

$$\sigma(\varepsilon) = \sigma_0 + K_v \left(1 - e^{-n_v \varepsilon} \right) \quad (15)$$

Using (3) and (6) it follows that:

$$\tau(\Gamma) = \tau_0 + K'_v \left(1 - e^{-n'_v \Gamma} \right) \quad (16)$$

$$\tau_0 = \frac{\sigma_0}{z} \quad (17)$$

$$K'_v = \frac{K_v}{z} \quad (18)$$

$$n'_v = \frac{n_v}{z} \quad (19)$$

The hardening coefficient H is given by:

$$H(\Gamma) = \frac{d\tau}{d\Gamma} = K'_v n'_v e^{-n'_v \Gamma} \quad (20)$$

Swift law adapted for yield elongation, with four parameters K (or σ_0), ε^* , ε_0 and n

$$\begin{aligned}\sigma(\varepsilon) &= \sigma_0 = K\varepsilon_0^n & \text{if } \varepsilon \leq \varepsilon^* \\ \sigma(\varepsilon) &= K(\varepsilon - \varepsilon^* + \varepsilon_0)^n = \sigma_0 \left(1 + \frac{\varepsilon - \varepsilon^*}{\varepsilon_0}\right)^n & \text{if } \varepsilon \geq \varepsilon^*\end{aligned}\quad (21)$$

Using (3) and (6) it follows that:

$$\begin{aligned}\tau(\Gamma) &= \tau_0 = K'\Gamma_0^n & \text{if } \Gamma \leq \Gamma^* \\ \tau(\Gamma) &= K'(\Gamma - \Gamma^* + \Gamma_0)^n = \tau_0 \left(1 + \frac{\Gamma - \Gamma^*}{\Gamma_0}\right)^n & \text{if } \Gamma \geq \Gamma^*\end{aligned}\quad (22)$$

$$K' = \frac{K}{z^{1+n}} \quad (23)$$

$$\tau_0 = \frac{\sigma_0}{z} \quad (24)$$

$$\Gamma_0 = z\varepsilon_0 \quad (25)$$

$$\Gamma^* = z\varepsilon^* \quad (26)$$

The hardening coefficient H is then given by:

$$\begin{aligned}H(\Gamma) &= \frac{d\tau}{d\Gamma} = 0 & \text{if } \Gamma \leq \Gamma^* \\ H(\Gamma) &= \frac{d\tau}{d\Gamma} = K'n(\Gamma - \Gamma^* + \Gamma_0)^{n-1} = \frac{\tau_0 n}{\Gamma_0} \left(1 + \frac{\Gamma - \Gamma^*}{\Gamma_0}\right)^{n-1} & \text{if } \Gamma \geq \Gamma^*\end{aligned}\quad (27)$$